

Electric Circuit

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Resonance frequency

The resonance will happen when

$$X_L = X_C$$

$$2\pi fL = \frac{1}{2\pi fC}$$

$$4\pi^2 f^2 LC = 1 \quad \therefore 4\pi^2 LC$$

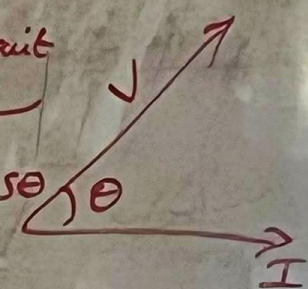
$$f^2 = \frac{1}{4\pi^2 LC} \quad \sqrt{}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

Resonance
frequency

$$Q\text{-factor} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Power in A.C. Circuit

$$P = VI \cos \theta$$


efficiency η

$$= \frac{P_{out}}{P_{in}}$$

$$h.p. = \text{horse power} = 746 \text{ watt}$$

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Example (1):

A series circuit contains a resistance of 4Ω and inductance of 0.5 H and a variable capacitance across 100 V 50 Hz supply. Find

(a) The capacitance for getting resonance

(b) The p.d. across inductance and capacitance.

(c) The Q-factor.

$$R = 4\Omega \quad L = 0.5\text{ H} \quad C = ?$$

$$V = 100\text{ V} \quad f = 50\text{ Hz}$$

(i) $C = ?$ (ii) p.d. V_L & $V_C = ?$ (iii) Q-factor = ?

(i) At resonance

$$X_L = X_C$$

$$2\pi f L = X_C$$

$$2\pi \times 50 \times 0.5 = X_C$$

$$X_C = 157.07\Omega$$

$$X_C = \frac{1}{2\pi f C}$$

$$\frac{1}{2\pi f C} = 157.07$$

$$C = \frac{1}{2\pi f \times 157.07}$$

$$= \frac{1}{2\pi \times 50 \times 157.07}$$

$$C = 2.026 \times 10^{-6}\text{ F}$$

$$C = 20.26 \times 10^{-8}\text{ F}$$

$$C = 20.26\mu\text{F}$$

$$\text{(ii)} V_L = V_C = I X_L = I X_C$$

At resonance

$$Z = R$$

and the current will be maximum

$$I = \frac{V}{R} = \frac{100}{4} = 25\text{ A}$$

$$\text{p.d.} = V_L = V_C = I X_L = 25 \times 157.07$$

$$\text{P.d.} = 3926.75\text{ V}$$

$$\text{(iii) Q-factor} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{4} \sqrt{\frac{0.5}{20.26 \times 10^{-6}}} = 39.274$$

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Example (2):

A 200 V / 50 Hz supply inductive circuit takes a current of 10 A, lagging 30° .

Find (i) resistance

(ii) reactance

(iii) inductance.

Solution:

$$V = 200 \text{ V} \quad f = 50 \text{ Hz}$$

$$I = 10 \text{ A} \quad \theta = 30^\circ$$

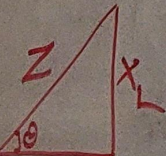
$$(i) R = ? \quad (ii) X_L = ? \quad (iii) L = ?$$

$$Z = \frac{V}{I} = \frac{200}{10} = 20 \Omega$$

$$\sin \theta = \frac{X_L}{Z}$$

$$X_L = Z \sin \theta$$

$$\text{and } = \frac{k}{Z} R$$



(i)

$$R = Z \cos \theta$$

$$R = 20 \cos 30^\circ$$
$$= 20 \times \frac{\sqrt{3}}{2}$$

$$R = 17.32 \Omega$$

$$X_L = Z \sin \theta$$

$$= 20 \sin 30^\circ$$

$$X_L = 10 \Omega$$

$$X_L = 2\pi f L$$

$$L = \frac{X_L}{2\pi f}$$

$$L = \frac{10}{2\pi \times 50}$$

$$L = 0.031 \text{ H}$$

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Example 3:

A single phase, 230 V, 50 Hz supply,
draws the power 2 h.p. from a.c. motor
of 12.5 A and power factor of 0.7.
What is the input power and
efficiency.

$$V = 230 \text{ V} \quad f = 50 \text{ Hz}$$

$$P = 2 \text{ h.p.} = 2 \times 746 = 1492 \text{ watt}$$

$$(i) P = V I \cos \theta$$

$$\text{P.f.} = \cos \theta$$

$$P_{in} = 230 \times 12.5 \times 0.7 = 2012.5 \text{ watt}$$

$$(ii) \text{ efficiency } \eta = \frac{P_{out}}{P_{in}}$$

$$\eta = \frac{1492}{2012.5}$$

$$\eta = 0.74136$$

$$\% \eta = 0.74136 \times 100$$

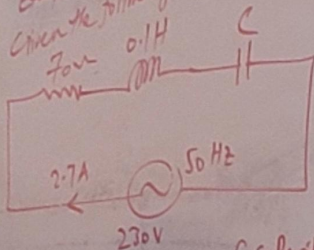
$$\boxed{\eta = 74.136 \%}$$

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Example 4:

Given the following R-L-C Series Circuit



Calculate the value of Capacitance

$$Z = \frac{V}{I} = \frac{230}{2.7} = 85.185 \Omega$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.1 = 31.415 \Omega$$

squaring both sides

$$Z^2 = R^2 + (X_L - X_C)^2$$

$$(85.185)^2 = (70)^2 + (31.415 - X_C)^2$$

$$7256.484 = 4900 + (31.415 - X_C)^2$$

$$7256.484 - 4900 = (31.415 - X_C)^2$$

$$2356.484 = (31.415 - X_C)^2$$

$$48.54 = 31.415 - X_C$$

$$48.54 - 31.415 = -X_C$$

$$17.125 = X_C$$

$$X_C = \frac{1}{2\pi fC}$$

$$C = \frac{1}{2\pi f X_C}$$

$$C = \frac{1}{2\pi \times 50 \times 17.125}$$

$$C = 1.858 \times 10^{-4} F$$

$$C = 185.8 \times 10^{-6} F$$

$$C = 185.8 \mu F$$

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Example (5):

A coil having a resistance of $10\ \Omega$ and inductance of $0.2\ \text{H}$ is connected to a $100\ \text{V}/50\ \text{Hz}$ supply. Calculate:

(a) the impedance

(b) reactance

(c) current

(d) the phase difference between current and voltage.

Solution:

$$R = 10\ \Omega \quad L = 0.2\ \text{H}$$

$$V = 100\ \text{V} \quad f = 50\ \text{Hz}$$

$$(a) Z = ? \quad (b) X_L = ? \quad (c) I = ? \quad (d) \theta = ?$$

$$(a) Z = \sqrt{R^2 + X_L^2}$$

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.2$$

$$X_L = 62.831\ \Omega$$

$$Z = \sqrt{(10)^2 + (62.831)^2}$$

$$Z = 63.621\ \Omega$$

$$(b) X_L = 62.831$$

$$(c) I = \frac{V}{Z} = \frac{100}{63.621}$$

$$I = 1.571\ \text{A}$$

$$(d) \theta = \tan^{-1} \left(\frac{X_L}{R} \right)$$

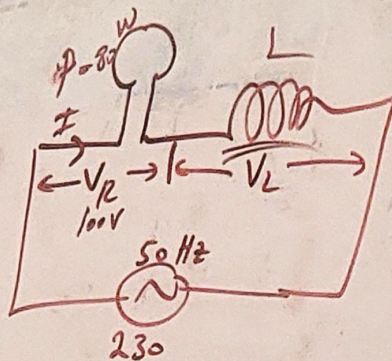
$$= \tan^{-1} \left(\frac{62.831}{10} \right)$$

$$\theta = 80.956^\circ$$

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Examp/c(7)

A 100V, 80W lamp is to be operated on 230V, 50 Hz supply. Calculate the inductance of the choke required to be connected in series with the lamp for this separation. The lamp can be taken as equivalent to a non-inductive resistance.



$$V = \sqrt{V_R^2 + V_L^2}$$

Squaring both side:

$$V^2 = V_R^2 + V_L^2$$

$$(230)^2 = (100)^2 + V_L^2$$

$$V_L^2 = (230)^2 - (100)^2 = 42900$$

$$V_L = 207.123 \text{ V}$$

$$P = VI$$

$$I = \frac{P}{V} = \frac{80}{100} = 0.8 \text{ A}$$

$$I = 0.8 \text{ A}$$

$$P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P} = \frac{(100)^2}{80}$$

$$R = 125 \Omega$$

$$Z = \sqrt{R^2 + X_L^2}$$

$$Z^2 = R^2 + X_L^2$$

$$X_L^2 = Z^2 - R^2$$

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$$Z = \frac{V}{I} = \frac{230}{0.8}$$

$$Z = 287.5 \Omega$$

$$X_L^2 = (287.5)^2 - (125)^2$$

$$= 67031.25$$

$$X_L = 258.903 \Omega$$

$$X_L = 2\pi f L$$

$$L = \frac{X_L}{2\pi f}$$

$$L = \frac{258.903}{2\pi \times 50}$$

$$L = 0.412 \text{ H}$$